

- 2 (a) State an example of how the motion of an object can be undergoing the following situations:

- (i) A body is instantaneously at rest but has a non-zero acceleration.

An object at its highest point when thrown vertically upwards on Earth or an object at the position of maximum displacement in a simple harmonic motion or any reasonable example.

..... [1]

- (ii) A body is moving with a constant speed but has a non-zero acceleration.

Any reasonable example of an object undergoing uniform circular motion.

.....

..... [1]

- (b) A toy car, initially at rest, can be made to propel forward by attaching a bottle filled with baking soda and vinegar. When the bottle cap is removed, the gas released from the reaction will cause the toy car to move forward. Fig. 2.1 shows how the acceleration, a , of the toy car changes with time t .

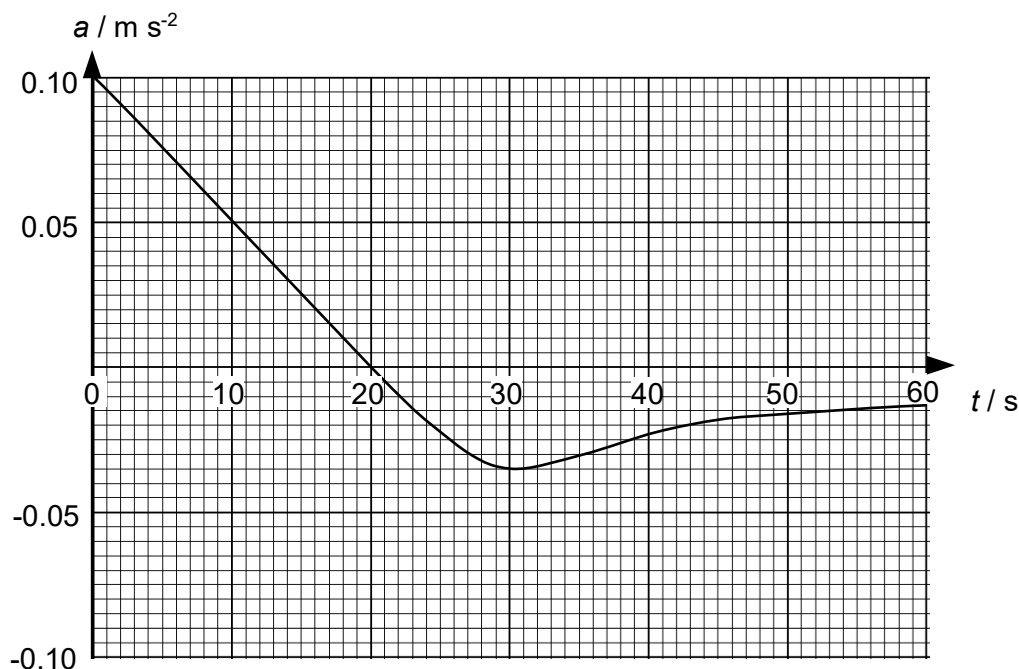


Fig. 2.1

- (i) With reference to Fig. 2.1, determine the maximum speed of the toy car.

Maximum speed occur at 20.0 s.

Maximum speed = Area under graph from 0 – 20.0 s

$$= \frac{1}{2} (0.100)(20.0)$$

$$= 1.00 \text{ m s}^{-1}$$

..... [1]

- (ii) Hence, sketch a graph on Fig. 2.2 to show how the velocity, v , of the toy car changes with time t .

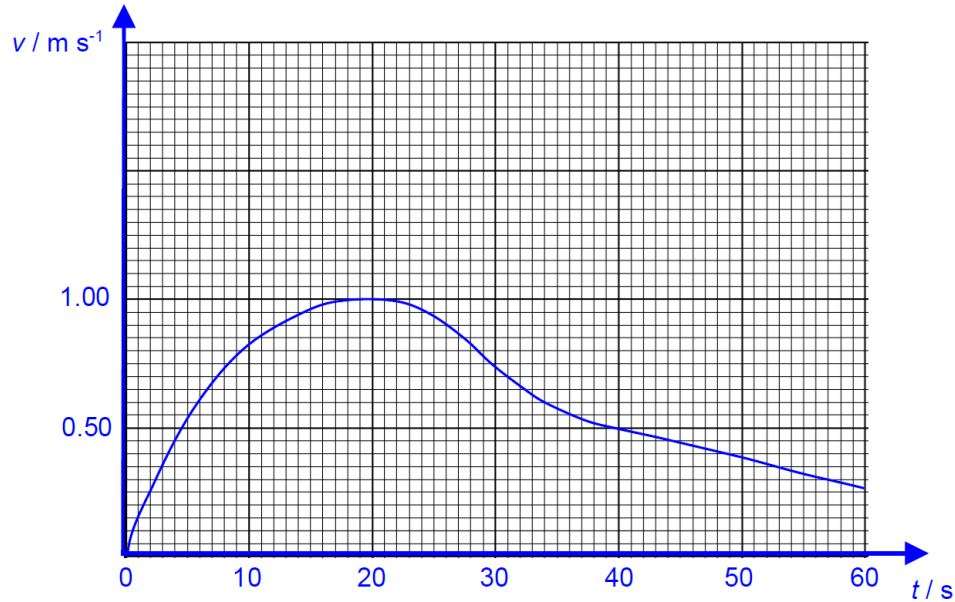


Fig. 2.2

[3]

[Total: 6]